

34-1e The Zero Lower Bound

As we have just seen, monetary policy works through interest rates. This conclusion raises a question: What if the Fed's target interest rate has fallen as far as it can? In the Great Recession of 2008 and 2009, the federal funds rate fell to about zero. In this situation, what, if anything, can monetary policy do to stimulate the economy?

Some economists describe this situation as a *liquidity trap*. According to the theory of liquidity preference, expansionary monetary policy works by reducing interest rates and stimulating investment spending. But if interest rates have already fallen to around zero, monetary policy may no longer be effective. Nominal interest rates cannot fall much below zero: Rather than making a loan at a negative nominal interest rate, a person would just hold cash. In this environment, expansionary monetary policy raises the supply of money, making the public's asset portfolio more liquid, but because interest rates can't fall any further, the extra liquidity might not have any effect. Aggregate demand, production, and employment may be "trapped" at low levels.

Other economists are skeptical about the relevance of liquidity traps and believe that a central bank continues to have tools to expand the economy, even after its interest rate target hits its lower bound of zero. One option is to have the central bank commit itself to keeping interest rates low for an extended period of time. Such a policy is sometimes called *forward guidance*. Even if the central bank's current target for the interest rate cannot fall any further, the promise that interest rates will remain low may help stimulate investment spending.

A second option is to have the central bank conduct expansionary open-market operations using a larger variety of financial instruments. Normally, the Fed conducts expansionary open-market operations by buying short-term government bonds. But it could also buy mortgage-backed securities and longer-term government bonds to lower the interest rates on these kinds of loans. This type of unconventional monetary policy is sometimes called *quantitative easing* because it increases the quantity of bank reserves. During the Great Recession, the Fed engaged in both forward guidance and quantitative easing.

Some economists have suggested that the possibility of hitting the zero lower bound for interest rates justifies setting the target rate of inflation well above zero. Under zero inflation, the real interest rate, like the nominal interest rate, can never fall below zero. But if the normal rate of inflation is, say, 4 percent, then the central bank can easily push the real interest rate to negative 4 percent by lowering the nominal interest rate to zero. Thus, a higher inflation target gives monetary policymakers more room to stimulate the economy when needed, reducing the risk of hitting the zero lower bound and having the economy fall into a liquidity trap.

QuickQuiz

1. According to the theory of liquidity preference, an economy's interest rate adjusts
 - a. to balance the supply and demand for loanable funds.
 - b. to balance the supply and demand for money.
 - c. one-for-one to changes in expected inflation.
 - d. to equal the interest rate prevailing in world financial markets.

2. If the central bank wants to contract aggregate demand, it can ____ the money supply and thereby ____ the interest rate.
 - a. increase; increase
 - b. increase; decrease
 - c. decrease; increase
 - d. decrease; decrease

3. The Fed's target for the federal funds rate
 - a. is an extra policy tool for the central bank, in addition to and independent of the money supply.
 - b. commits the Fed to set a particular money supply so that it hits the announced target.
 - c. is a goal that is rarely achieved because the Fed can determine only the money supply.
 - d. matters to banks that borrow and lend federal funds but does not influence aggregate demand.

